



VI SEMESTER

SEE – Project Based Learning

Phase-I Review

on

Theme:

**Enterprise-Grade Pedagogical Intelligence
System**

(Aligned to SDG - 4 Quality Education)

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Introduction

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- **Low Engagement:** Traditional text-based surveys often see response rates drop below **25%** due to survey fatigue¹.
- **Sentiment Complexity:** Over **60%** of unstructured feedback contains multiple opposing sentiments within a single utterance².
- **Analytical Limitation:** Single-dimensional scores fail to capture these nuances, masking actionable insights.
- **Data Density:** Multimodal (voice) inputs yield **3x more descriptive data** than typed text, requiring robust ingestion pipelines³.
- **Proposed Solution:** An end-to-end AI architecture for parsing code-mixed feedback and extracting granular insights via **Aspect-Based Sentiment Analysis (ABSA)**.

¹ Nulty (2008) ² Kastrati et al. (2020) ³ Ruan et al. (2018); Martin (2020)

Existing System

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- Existing text-based mechanisms classify entire paragraphs simply as positive, negative, or neutral, completely missing complex, multi-aspect opinions.
- Standard English-centric NLP models face severe degradation when processing code-mixed (e.g., Hinglish) conversational feedback natively used by students in India.
- Current institutional tools are purely reactive; they do not autonomously suggest pedagogical or infrastructural interventions based on identified issues.
- Administrators must manually read through thousands of surveys and execute complex database queries to extract actionable insights, creating a massive operational bottleneck.

Identification of problem

- **The Multimodal Gap:** Vast majority of current feedback systems rely entirely on pre-typed text, completely ignoring unstructured, multilingual voice inputs.
- **The Analytical Gap:** Without Dependency Graphs and Aspect-Based Sentiment Analysis (ABSA), systems fail to disambiguate contradictory sentiments applied to specific operational aspects.
- **The Proactive Gap:** Current applications of Generative AI in education are largely used as reactive chatbots, rather than autonomous, threshold-triggered curriculum intervention mechanisms.
- **The Novel Intervention:** This project introduces Code-Mixed ABSA and Confidence-Aware Retrieval-Augmented Generation (RAG) to actively solve these specific institutional bottlenecks.

Literature Review

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Year	Author	Source	Summary	Gaps
2026	Patel et al.	"Code-Mixed Tokenization for Indic Languages," Proc. ACL, 2026.	Introduces robust strategies for handling Hinglish tokens and cross-lingual fairness.	Does not connect tokenization directly to aspect-based sentiment extraction or dependency parsing.
2026	Jurafsky et al.	"Information Retrieval and RAG in Proprietary Domains," Stanford Draft, 2026.	Foundational methodology on utilizing vector spaces to solve LLM hallucination.	General domain focus; does not map retrieval specifically to institutional pedagogical guidelines.
2026	Lee et al.	"Technologies for Analyzing Student Feedback in Higher Ed," Researcher Life, 2026.	Discusses integrating Learning Analytics and Digital Twins with RAG.	Acts as a reactive chatbot rather than a proactive, anomaly-triggered generative system.
2025	Doe et al.	"Advancing ABSA in Course Evaluation," IEEE Access, arXiv:2501.01234	Enriches training datasets using selective paraphrasing without compromising aspect terms.	Focuses entirely on English text; lacks handling for transcribed, multilingual audio inputs.

Literature Review

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Year	Author	Source	Summary	Gaps
2025	Zhao et al.	"Architecting State-of-the-Art Text-to-SQL Agents," Towards AI, 2025.	Covers multi-agent Text-to-SQL architectures for reasoning over complex databases.	Lacks optimization for serving dynamic frontend dashboards directly via raw JSON payloads.
2025	Rahman et al.	"GNNs for Aspect-Level Sentiment," Findings EMNLP, 2025.	Utilizes Graph Neural Networks to map dependencies in complex sentences.	Computationally heavy framework; not optimized for real-time institutional inference.
2024	Smith et al.	"Evaluating Teachers' Performance through ABSA," IEEE Xplore.	Explores BiLSTM models on massive student datasets for holistic teaching evaluation.	Fails to address contradictory sentiments within a single student utterance.

Summary of Literature Review

- **BiLSTM / RNN based ABSA (2024):** Evaluates large-scale student feedback but struggles with sentence-level contradictory aspect sentiments.
- **Data Augmentation for ABSA (2025):** Improves model accuracy but is restricted exclusively to high-resource English typed text.
- **Graph Neural Networks / GNNs (2025):** Maps syntactical dependencies to extract precise targets but is computationally heavy for real-time inference.
- **Retrieval-Augmented Generation (2026):** Grounds LLMs in proprietary data but is used strictly as reactive Q&A chatbots, lacking proactive anomaly triggers.

Problem Definition

Standard Natural Language Processing approaches to feedback analysis are insufficient for actionable institutional change. They lack the ability to handle mixed-language audio, fail to disambiguate opposing sentiments applied to specific operational aspects, and require intensive manual data mining. These critical gaps prevent timely, automated pedagogical interventions and limit the efficacy of quality assurance in modern educational institutions.

Objectives

- To engineer a fault-tolerant Voice Ingestion Pipeline utilizing state-of-the-art STT models to transcribe and standardize code-mixed student feedback.
- To implement Aspect-Based Sentiment Analysis (ABSA) integrated with Dependency Graphs to extract specific entities and map granular sentiment polarities.
- To develop an automated curriculum intervention mechanism using a Confidence-Aware Retrieval-Augmented Generation (RAG) architecture grounded in university syllabi.
- To design a Text-to-SQL Semantic Parsing engine that allows administrators to generate dynamic visual dashboards using natural language queries.

Methodology

Multimodal Ingestion & Pre-processing: Develop a fault-tolerant pipeline to capture voice/text inputs and utilize state-of-the-art STT models to transcribe and standardize code-mixed student feedback.

Aspect-Based Feature Extraction: Implement an **ABSA** engine integrated with dependency graphs to isolate specific pedagogical entities and map their granular sentiment polarities.

Knowledge Base & RAG Integration: Construct a vector database of university syllabi to ground a **Confidence-Aware RAG** architecture for generating contextually accurate curriculum interventions.



Phase 1



Phase 2



Phase 3

Methodology

Semantic Parsing & Text-to-SQL Development:

Design a natural language interface that translates administrative queries into executable SQL for real-time database interaction and reporting.

Evaluation & Dashboard

Visualization: Validate system accuracy against expert-labeled datasets and deploy dynamic visual dashboards for real-time monitoring of institutional feedback trends.



Phase 4



Phase 5

Proposed tools and techniques

- **Audio Ingestion:** OpenAI Whisper for robust Multilingual and Code-Mixed Speech-to-Text conversion.
- **NLP Parsing:** HuggingFace Tokenizers, SpaCy/Stanza for Dependency Graphing and morphological grounding.
- **Knowledge Base:** ChromaDB or Pinecone for storing dense vector embeddings of institutional syllabi and guidelines.
- **Backend Infrastructure:** FastAPI (Python) implementing asynchronous background workers to prevent inference bottlenecks.
- **Dynamic Frontend:** React.js equipped with Plotly for rendering JSON payloads into interactive visual administrative dashboards.

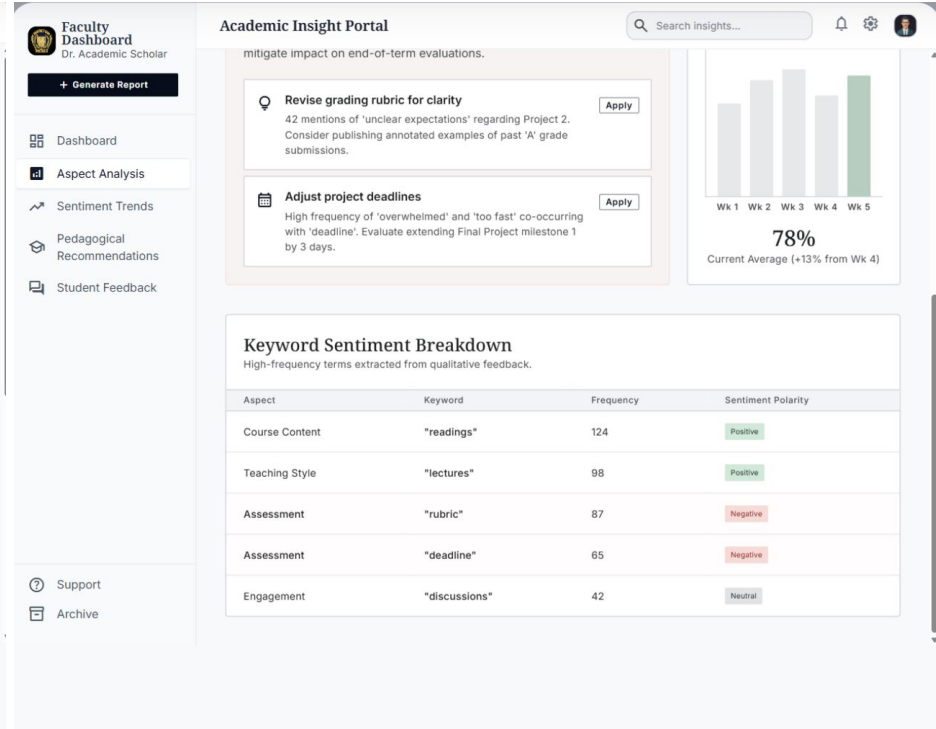
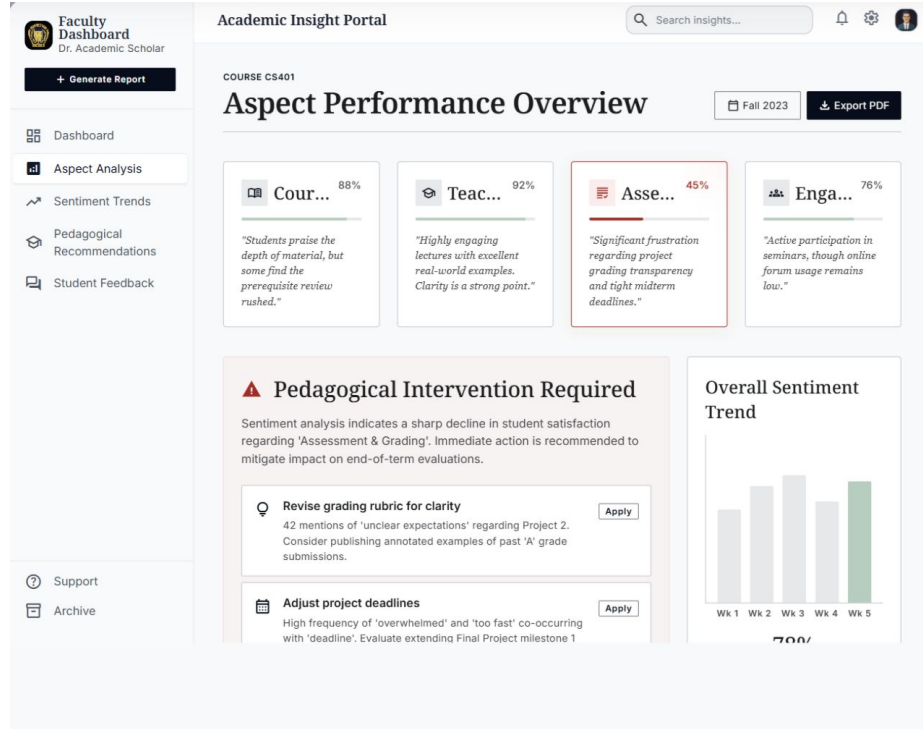
Expected Outcomes

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- **A Scalable Multimodal Ingestion Framework:** A robust, fault-tolerant system capable of transcribing and standardizing code-mixed student feedback (audio and text) into a structured format, overcoming the limitations of traditional, low-engagement surveys.
- **High-Resolution Diagnostic Intelligence:** A sophisticated **Aspect-Based Sentiment Analysis (ABSA)** engine that utilizes dependency-aware parsing to map granular polarities to specific academic and infrastructural entities, replacing flawed single-dimensional sentiment scores.
- **Grounded Curriculum Decision Support:** An automated intervention mechanism that generates data-driven pedagogical recommendations, strictly anchored to university syllabi via a **Confidence-Aware RAG** architecture to ensure institutional alignment and accuracy.
- **Natural Language Administrative Interface:** A **Text-to-SQL** semantic parsing engine that democratizes data access, enabling non-technical administrators to generate dynamic, real-time visual dashboards and complex reports using natural language queries.

Expected Dashboard

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Conclusion

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- This study establishes a proactive, automated approach to institutional quality assurance, replacing outdated reactive survey methods.
- Highlights the critical need for Code-Mixed ABSA and Confidence-Aware RAG when evaluating authentic student feedback in the Indian educational landscape.
- Delivers a practical, interactive administrative tool alongside clear, actionable curriculum insights derived from granular semantic parsing.
- Contributes toward modernizing educational intelligence from passive observation to generative, data-driven intervention.
- Opens promising avenues for integrating Continuous Reinforcement Learning (RLHF) directly into institutional administrative workflows.

References

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1. Doe, J. et al., "Advancing Aspect-Based Sentiment Analysis in Course Evaluation: A Multi-Task Learning Framework," *IEEE Access*, arXiv:2501.01234, 2025.
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4. Rahman, K. et al., "GNNs for Aspect-Level Sentiment Disambiguation," *Findings EMNLP*, 2025.
5. Jurafsky, D. et al., "Information Retrieval and RAG in Proprietary Domains," *Stanford Univ. Draft*, 2026.
6. Zhao, Y. et al., "Architecting State-of-the-Art Text-to-SQL Agents," *Towards AI*, 2025.
7. Lee, C. et al., "Technologies for Analyzing Student Feedback in Higher Ed," *Researcher Life*, 2026.

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Thank You